

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

LAB 4 REPORT

SMTP Email Traffic Forensics

Course Name	CIP-B103: Network Forensics Fundamentals
Lab Title	CIP-B103 Lab 4: SMTP Email Traffic Forensics
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2. Executive Summary

This lab conducted a forensic examination of a historical SMTP packet capture (smtp.pcap). The primary objectives were to reassemble TCP streams, extract network and application metadata, identify authentication mechanisms, decode Base64 parameters offline, and reconstruct message content without compromising sensitive data. Analysis confirmed plain-text SMTP communication occurring over default port 25. Plain-text Base64 parameters yielded user credentials, and stream reassembly recovered a multi-part email sent via Microsoft Outlook 12.0 containing text body parts and a plain-text file attachment (NEWS.txt).

3. Lab Objectives

- Distinguish plaintext SMTP, STARTTLS, and implicit TLS mechanisms.
- Identify SMTP client-server command/response sequences and authentication mechanisms.
- Reassemble TCP streams to extract message headers, body parts, and attachments.
- Perform secure offline Base64 decoding of authentication parameters.
- Document client/server network identifiers, IP/MAC addresses, and software user agents.
- Evaluate evidential limitations when handling unencrypted or partially captured traffic.

4. Tools and Resources Used

- **Wireshark (v4.x):** Graphical packet analyzer used to inspect SMTP flags, reconstruct TCP streams, and analyze MIME encapsulation.
- **tshark:** Command-line packet capture and analysis tool used for automated packet field extraction and reporting.
- **Python 3 (base64 module):** Used for controlled, offline decoding of captured Base64 strings.
- **sha256sum & capinfos:** Command-line utilities used for cryptographic verification and capture file metadata generation.
- **Evidence File:** smtp.pcap sourced from Wireshark Sample Captures.

5. Methodology

5.1 Evidence Integrity & Environment Preparation

1. Created the lab directory structure and verified folder creation:

```
(root@icdfa)-[~]  
# mkdir -p ~/CIP-B103-Lab4/{evidence,working,exported,reports,screenshots,scripts}  
cd ~/CIP-B103-Lab4
```

2. Downloaded smtp.pcap into evidence/, duplicated it to working/smtp_working.pcap preserving timestamps, and generated SHA-256 hashes to establish chain-of-custody:

```

(root@icdfa) [~/CIP-B103-Lab4]
# wget -O evidence/smtp.pcap 'https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap'
cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap
--2026-08-07 18:38:35-- https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org) ... 104.26.11.240, 172.67.75.39, 104.26.10.240, ...
Connecting to wiki.wireshark.org (wiki.wireshark.org)|104.26.11.240|:443 ... connected.
HTTP request sent, awaiting response ... 302 Found
Location: https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap [following]
--2026-08-07 18:38:35-- https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
Resolving gitlab.com (gitlab.com) ... 172.65.251.78, 2606:4700:90:0:f22e:fbec:5bed:a9b9
Connecting to gitlab.com (gitlab.com)|172.65.251.78|:443 ... connected.
HTTP request sent, awaiting response ... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: 'evidence/smtp.pcap'

evidence/smtp.pcap      100%[=====] 27.20K  --.-KB/s   in 0.01s

2026-08-07 18:38:36 (1.93 MB/s) - 'evidence/smtp.pcap' saved [27850/27850]

(root@icdfa) [~/CIP-B103-Lab4]
# sha256sum evidence/smtp.pcap working/smtp_working.pcap | tee reports/smtp_capture_hashes.txt
capinfos evidence/smtp.pcap | tee reports/smtp_capinfos.txt
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab evidence/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab working/smtp_working.pcap
File name: evidence/smtp.pcap
File type: Wireshark/tcpdump/ ... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 60
File size: 27 kB
Data size: 26 kB
Capture duration: 9.198384 seconds
Earliest packet time: 2009-10-05 07:06:07.492060
Latest packet time: 2009-10-05 07:06:16.690444
Data byte rate: 2,920 bytes/s
Data bit rate: 23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256: 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1: 3def1a1fed849e7f23b66e6925ddff05845a400b
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 60

```

5.2 Locating and Inventorying SMTP Streams

Profiled TCP conversations and inventoried all SMTP frames:

```
tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt
tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
    -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info \
    | tee reports/smtp_packet_inventory.tsv
Running as user "root" and group "root". This could be dangerous.
```

```
TCP Conversations
Filter:<No Filter>
```

			←		→		Total	Relative	Duration
				Frames Bytes		Frames Bytes			
10.10.1.4:1470	↔	74.53.140.153:25		25 1,980 bytes		28 22 kB		53 24 kB	0.036986000 7.577

```
Running as user "root" and group "root". This could be dangerous.
```

No	Time	Source	Destination	Protocol	Length	Info
6	Oct 5, 2009 07:06:08.219663000	WAT	74.53.140.153 25	10.10.1.4 1470	S: 220<xc90.websitewelcome.com ESMTExim 4.69	
#1 Mon, 05 Oct 2009 01:05:54 -0500	We do not authorize the use of this system to transport unsolicited, and/or bulk e-mail.					
7	Oct 5, 2009 07:06:08.224809000	WAT	10.10.1.4 1470	74.53.140.153 25	C: EHLO GP	
9	Oct 5, 2009 07:06:08.566183000	WAT	74.53.140.153 25	10.10.1.4 1470	S: 250<xc90.websitewelcome.com Hello GP [122.16	
2.443.157]	SIZE 52428800 PIPELINING AUTH PLAIN LOGIN STARTTLS HELD					
10	Oct 5, 2009 07:06:08.568729000	WAT	10.10.1.4 1470	74.53.140.153 25	C: AUTH LOGIN	
11	Oct 5, 2009 07:06:08.911081000	WAT	74.53.140.153 25	10.10.1.4 1470	S: 334 VxNlcm5HbwU6	
12	Oct 5, 2009 07:06:08.911655000	WAT	10.10.1.4 1470	74.53.140.153 25	C: User: Z3VycGfydGFwQHBhdHJpb3RzMmLu	
13	Oct 5, 2009 07:06:09.253544000	WAT	74.53.140.153 25	10.10.1.4 1470	S: 334 UGFzc3dvcmQ6	
14	Oct 5, 2009 07:06:09.254118000	WAT	10.10.1.4 1470	74.53.140.153 25	C: Pass: cHVuamFIQDEyMw==	
15	Oct 5, 2009 07:06:09.613798000	WAT	74.53.140.153 25	10.10.1.4 1470	S: 235 Authentication succeeded	
16	Oct 5, 2009 07:06:09.614414000	WAT	10.10.1.4 1470	74.53.140.153 25	C: MAIL FROM: <gurpartap@patriots.in>	
17	Oct 5, 2009 07:06:09.956765000	WAT	74.53.140.153 25	10.10.1.4 1470	S: 250 OK	
18	Oct 5, 2009 07:06:09.957250000	WAT	10.10.1.4 1470	74.53.140.153 25	C: RCPT TO: <raj_deol2002in@yahoo.co.in>	
19	Oct 5, 2009 07:06:10.319708000	WAT	74.53.140.153 25	10.10.1.4 1470	S: 250 Accepted	
20	Oct 5, 2009 07:06:10.320803000	WAT	10.10.1.4 1470	74.53.140.153 25	C: DATA	
21	Oct 5, 2009 07:06:10.661679000	WAT	74.53.140.153 25	10.10.1.4 1470	S: 354 Enter message, ending with "." on a line	
by itself						
22	Oct 5, 2009 07:06:10.692743000	WAT	10.10.1.4 1470	74.53.140.153 25	C: DATA fragment, 1460 bytes	
23	Oct 5, 2009 07:06:10.692786000	WAT	10.10.1.4 1470	74.53.140.153 25	C: DATA fragment, 1460 bytes	
24	Oct 5, 2009 07:06:10.692804000	WAT	10.10.1.4 1470	74.53.140.153 25	C: DATA fragment, 1460 bytes	
25	Oct 5, 2009 07:06:10.692823000	WAT	10.10.1.4 1470	74.53.140.153 25	[TCP Window Full] C: DATA fragment, 1460 bytes	
26	Oct 5, 2009 07:06:10.695115000	WAT	192.168.1.1,10.10.1.4 1470	10.10.1.4,74.53.140.153 25	Destination unreachable (Fragme	
ntation needed)						
27	Oct 5, 2009 07:06:10.695115000	WAT	192.168.1.1,10.10.1.4 1470	10.10.1.4,74.53.140.153 25	Destination unreachable (Fragme	
ntation needed)						
28	Oct 5, 2009 07:06:10.695623000	WAT	192.168.1.1,10.10.1.4 1470	10.10.1.4,74.53.140.153 25	Destination unreachable (Fragme	
ntation needed)						
29	Oct 5, 2009 07:06:10.696248000	WAT	192.168.1.1,10.10.1.4 1470	10.10.1.4,74.53.140.153 25	Destination unreachable (Fragme	
ntation needed)						
30	Oct 5, 2009 07:06:10.696634000	WAT	192.168.1.1,10.10.1.4 1470	10.10.1.4,74.53.140.153 25	Destination unreachable (Fragme	
ntation needed)						
38	Oct 5, 2009 07:06:11.494181000	WAT	10.10.1.4 1470	74.53.140.153 25	C: DATA fragment, 1452 bytes	
39	Oct 5, 2009 07:06:11.494199000	WAT	10.10.1.4 1470	74.53.140.153 25	C: DATA fragment, 1452 bytes	
41	Oct 5, 2009 07:06:11.834628000	WAT	10.10.1.4 1470	74.53.140.153 25	C: DATA fragment, 1452 bytes	
42	Oct 5, 2009 07:06:11.834655000	WAT	10.10.1.4 1470	74.53.140.153 25	C: DATA fragment, 1452 bytes	
44	Oct 5, 2009 07:06:11.858316000	WAT	10.10.1.4 1470	74.53.140.153 25	C: DATA fragment, 1452 bytes	
45	Oct 5, 2009 07:06:11.858334000	WAT	10.10.1.4 1470	74.53.1		

5.3 Command and Response Analysis

Processed client commands (EHLO, AUTH LOGIN, MAIL FROM, RCPT TO, DATA, QUIT) and server response codes (220, 250, 334, 354, 235):

6	Oct	5,	2009	07:06:08.21966300	WAT	74.53.140.153	10.10.1.4	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2
7	001:05:54	-0500	We do not authorize the use of this system to transport unsolicited, and/or bulk e-mail.					
8	Oct	5,	2009	07:06:08.224889000	WAT	10.10.1.4	74.53.140.153	C: EHLO GP
9	Oct	5,	2009	07:06:08.254720000	WAT	74.53.140.153	10.10.1.4	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] SIZ
10	E 52428800	PIPELINING AUTH PLAIN LOGIN STARTTLS HELP						
11	Oct	5,	2009	07:06:08.568729000	WAT	10.10.1.4	74.53.140.153	C: AUTH LOGIN
12								
13	Oct	5,	2009	07:06:09.614414000	WAT	10.10.1.4	74.53.140.153	C: MAIL FROM: <gurpartap@patriots.in>
14	Oct	5,	2009	07:06:09.956765000	WAT	74.53.140.153	10.10.1.4	S: 250 OK
15	Oct	5,	2009	07:06:09.957250000	WAT	10.10.1.4	74.53.140.153	C: RCPT TO: <raj_deol2002in@yahoo.co.in>
16	Oct	5,	2009	07:06:10.319708000	WAT	74.53.140.153	10.10.1.4	S: 250 Accepted
17	Oct	5,	2009	07:06:10.320203000	WAT	10.10.1.4	74.53.140.153	C: DATA

5.4 Base64 Authentication Decoding

1. Extracted encoded Base64 strings from frames 12 and 14 (Z3VycGFydGFwQHBhdHJpb3RzLmlu and cHVuamFiQDEyMw==).
2. Executed an offline Python script to decode strings safely without third-party web tools:

```

(root@icdfa)-[~/CIP-B103-Lab4]
# python3 - <<'PYLAB'
import base64
samples = {
    'username': 'Z3VycGFydGFwQHBhdHJpb3RzLmlu',
    'password': 'cHVuamFiQDEyMw=='
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        # Masked output for safety as required by policy
        masked = decoded[0] + "*" * (len(decoded) - 2) + decoded[-1]
        print(f'{label} (Decoded & Masked): {masked}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
PYLAB
username (Decoded & Masked): g*****n
password (Decoded & Masked): p*****3

```

5.5 Message Reassembly & Header Extraction

Extracted raw stream and headers to verify MIME boundary structure (----=_NextPart_000_0004_01CA45B0.095693F0):

```
(root@icdfa) ~/CIP-B103-Lab4
# tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 | tee reports/smtp_stream_0.txt
grep -Ei '^(Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):' \
reports/smtp_stream_0.txt | tee reports/message_headers.txt
Running as user "root" and group "root". This could be dangerous.
```

```
Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
181
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.

9
EHLO GP

137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

12
AUTH LOGIN

18
334 VXNlcm5hbWU6

30
Z3VycGFydGFwQHhhdHJpb3RzLmlu

18
334 UGFzc3dvcmQ6
```

```
30
Z3VycGFydGFwQHBhdHJpb3RzLmlu

      18
334 UGFzc3dvcmQ6

18
cHVuamFiQDEyMw==

      30
235 Authentication succeeded

36
MAIL FROM: <gurpartap@patriots.in>

      8
250 OK

39
RCPT TO: <raj_deol2002in@yahoo.co.in>

      14
250 Accepted

6
DATA

      56
354 Enter message, ending with "." on a line by itself

1460
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
    .boundary="——=_NextPart_000_0004_01CA45B0.095693F0"
X-Mailer: Microsoft Office Outlook 12.0
```



```
<!--[if gte mso 9]><xml>
  <o:shapedefaults v:ext=3D"edit" spidmax=3D"1026" />
</xml><![endif]>--<!--[if gte mso 9]><xml>
  <o:shapelayout v:ext=3D"edit">
    <o:idmap v:ext=3D"edit" data=3D"1" />
  </o:shapelayout></xml><![endif]>-->
</head>

<body lang=3DEN-US link=3Dblue vlink=3Dpurple>

<div class=3DSection1>
1460

<p class=3DMsoNormal>Hello<o:p></o:p></p>

<p class=3DMsoNormal><o:p>&nbsp;</o:p></p>

<p class=3DMsoNormal>I send u smtp pcap file <o:p></o:p></p>

<p class=3DMsoNormal>Find the attachment<o:p></o:p></p>

<p class=3DMsoNormal><o:p>&nbsp;</o:p></p>

<p class=3DMsoNormal>GPS<o:p></o:p></p>

</div>

</body>

</html>

-----=_NextPart_001_0005_01CA45B0.095693F0--

-----=_NextPart_000_0004_01CA45B0.095693F0
Content-Type: text/plain;
.name="NEWS.txt"
Content-Transfer-Encoding: quoted-printable
Content-Disposition: attachment;
.filename="NEWS.txt"
```

```
Version 4.9.4.1 (5.0 beta 4.1):

* back to gcc 2.95.3
* Profiling support
* new update/packages checker (vUpdate)
* Lots of bugfixes

-----=_NextPart_000_00
29
04_01CA45B0.095693F0--

.

28
250 OK id=1Mugho-0003Dg-Un

6
QUIT

48
221 xc90.websitewelcome.com closing connection

=====
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;
```

6. Screenshots and Evidence

- **Figure 1:** SHA-256 hash output and capinfos execution summary.
- **Figure 2:** Wireshark frame list showing TCP 3-way handshake and initial SMTP 220 Service ready banner.
- **Figure 3:** Plaintext AUTH LOGIN exchange showing 334 Username: and 334 Password: challenges.
- **Figure 4:** Python offline terminal execution decoding Base64 credentials.
- **Figure 5:** Reassembled TCP Stream showing full MIME headers, email body text, and NEWS.txt attachment block.

7. Analysis and Findings

7.1 SMTP Command/Response Timeline Matrix

Seq	Direction	Command / Code	Parameter / Value	Forensic Interpretation	Timestamp / Relative Time
1	Server \$\\rightarrow\$ Client	220	xc90.websitewelcome.com ESMTP Exim 4.69 #1	SMTP Service Ready Banner	Oct 5, 2009 07:06:08.2 196 WAT
2	Client \$\\rightarrow\$ Server	EHLO	GP	Client Greeting	Oct 5, 2009 07:06:08.2 248 WAT
3	Server \$\\rightarrow\$ Client	250	xc90.websitewelcome.com Hello GP... (Advertises STARTTLS)	Server Capabilities List	Oct 5, 2009 07:06:08.5 661 WAT
4	Client \$\\rightarrow\$ Server	AUTH LOGIN	N/A	Authentication Request	Oct 5, 2009 07:06:08.5 687 WAT
5	Server \$\\rightarrow\$ Client	334	VXNlcm5hbWU6 (Username:)	Username Challenge	Oct 5, 2009 07:06:08.9 110 WAT
6	Client \$\\rightarrow\$ Server	Response	Z3VycGFydGFwQHBhdHJpb3RzLmlu	Encoded Username	Oct 5, 2009 07:06:08.9 116 WAT
7	Server \$\\rightarrow\$ Client	334	UGFzc3dvcmQ6 (Password:)	Password Challenge	Oct 5, 2009 07:06:09.2 535 WAT
8	Client	Response	Y2hWdWFtRmlRREV	Encoded	Oct 5, 2009

	\$\rightarrow\$ w\$ Server		5TXc==	Password	07:06:09.2 541 WAT
9	Server \$\rightarrow\$ w\$ Client	235	Authentication succeeded	Auth Granted	Oct 5, 2009 07:06:09.6 137 WAT
10	Client \$\rightarrow\$ w\$ Server	MAIL FROM	<u><gurpartap@patriot s.in></u>	Envelope Sender	Oct 5, 2009 07:06:09.6 144 WAT
11	Server \$\rightarrow\$ w\$ Client	250	OK	Sender Accepted	Oct 5, 2009 07:06:09.9 567 WAT
12	Client \$\rightarrow\$ w\$ Server	RCPT TO	<u><raj_deol2002in@y ahoo.co.in></u>	Envelope Recipient	Oct 5, 2009 07:06:09.9 572 WAT
13	Server \$\rightarrow\$ w\$ Client	250	Accepted	Recipient Accepted	Oct 5, 2009 07:06:10.3 197 WAT
14	Client \$\rightarrow\$ w\$ Server	DATA	N/A	Start of Email Payload	Oct 5, 2009 07:06:10.3 202 WAT
15	Server \$\rightarrow\$ w\$ Client	354	Enter message, ending with '.'	Ready for Payload	Oct 5, 2009 07:06:10.6 616 WAT
16	Client \$\rightarrow\$ w\$ Server	QUIT	N/A	Terminate Session	Oct 5, 2009 07:06:14.7 638 WAT

7.2 Base64 Authentication Recovery

- **Captured Encoded User:** Z3VycGFydGFwQHBhdHJpb3RzLmlu -> Decoded: gurpartap@patriots.in (Redacted in public text: g*****p@patriots.in)
- **Captured Encoded Pass:** cHVuamFiQDEyMw== -> Decoded: punjab@123 (Redacted in public text: p*****3)

7.3 Email Message Findings & Metadata Summary

- **Session Timestamps:** Mon, 05 Oct 2009 07:06:07.4920 WAT to 07:06:16.6904 WAT (Total Capture Duration: 9.1983s).
- **Client Identifiers:** IP Address: 10.10.1.4 | Ephemeral TCP Port: 1470 | MAC Address: 00:e0:1c:3c:17:c2.
- **Server Identifiers:** IP Address: 74.53.140.153 | Standard SMTP TCP Port: 25 | MAC Address: 00:1f:33:d9:81:60.
- **Client Software / User-Agent:** Microsoft Office Outlook 12.0 (X-Mailer: Microsoft Office Outlook 12.0).
- **Header Data:** From: "Gurpartap Singh" <gurpartap@patriots.in> | To: <raj_deol2002in@yahoo.co.in> | Date: Mon, 5 Oct 2009 11:36:07 +0530 | Subject: SMTP | Message-ID: <000301ca4581\$ef9e57f0\$cedb07d0\$@in>
- **Body Content & Attachment:** Multi-part MIME (multipart/mixed containing multipart/alternative).
- **Plaintext Body:** *"Hello \r\n I send u smtp pcap file \r\n Find the attachment \r\n GPS"*
- **Attachment:** Encapsulated plain-text file named NEWS.txt (114 lines of software changelog/release notes starting with Version 4.9.4.1 (5.0 beta 4.1):).

7.4 Encryption Assessment & Evidential Limitations

The session operated purely in plain-text over default port 25. Although the server advertised 250-STARTTLS in its EHLO response capability list, the client did not issue a STARTTLS command. Consequently, no TLS handshake occurred, leaving all application headers, credentials, and payload contents fully exposed in cleartext.

8. Challenges and Solutions

- **Challenge:** The password i had was different from the given one in the lab.
- **Solution:** I changed the password to what i discovered in my work compared to what was given in the lab.

9. Conclusion

This lab demonstrated how unencrypted legacy protocols like standard SMTP expose sensitive authentication details and communications to passive network monitoring. Base64 encoding provides no security or privacy, serving only as a transport formatting mechanism.

10. Recommendations

- Enforce mandatory TLS encryption (STARTTLS on port 587 or implicit SMTPS on port 465) across mail transfer agents.
- Disable plain-text authentication methods (AUTH LOGIN / AUTH PLAIN) over unencrypted port 25 connections.

11. References

- Wireshark Foundation. SampleCaptures: smtp.pcap. Retrieved from <https://wiki.wireshark.org/SampleCaptures>
- RFC 5321: Simple Mail Transfer Protocol. IETF.